MSE 311 Thermodynamics of materials-Homework 13

Name: *Student ID:*

1. The probability that a drunk man moves the distance z by n steps is

$$P(n,z) \approx \frac{1}{\sqrt{2\pi n\lambda^2}} \exp\left(-\frac{z^2}{2n\lambda^2}\right)$$

What type of distribution function does this distribution belong to?

2. In a γ-Fe, lattice parameter = 0.37 nm. At 1000°C, $D = 2.5 \times 10^{-11}$ $m^2\ s^{-1}$, what the jumping frequency v_B?

3. A certain amount of element B was plated on one end of a slender metal rod, and the concentration of element B at 10 μm, 20 μm, 30 μm, 40 μm and 50 μm away from the plating end was 83.8 g/m^3, 66.4 g/m^3, 42.0 g/m^3, 23.6 g/m^3 and 8.74 g/m^3, respectively, after 24 hours of high temperature annealing. Please estimate simply the diffusion coefficient of B in this metal.

4. When a weight was hung on a pure Cu wire of radius r = 0.1 mm and held at a high temperature, the balanced load of the weight was $m = 5.5\times10^{-5}$ kg, which caused neither expansion nor shrinkage of the wire. Estimate the surface energy of pure Cu from this experimental value.

5. Steel can be hardened and made more corrosion resistant by heat treating in a nitrogen atmosphere to diffuse nitrogen into the surface laver of the steel. This process can be roughly modeled using error function solution as following.

$$C(y,t) = A + B\text{erf}\left(\frac{y}{\sqrt{4Dt}}\right)$$

Assume that a plate of steel is heat treated in a nitrogen environment at 500°C, where the solubility of nitrogen in steel is $C_N^* \approx 0.5$ mol % and the initial background concentration of nitrogen in the steel is zero ($C_N^0 = 0$).

(a) Provide the boundary condition and initial condition for this problem.

(b) Based on your boundary and initial conditions, provide the *A* and *B* of error function solution in this problem.

(c) Assuming the diffusivity of nitrogen in steel is $D_{\text{N/steel}} = 10^{-9}$ cm^2/s at the heat treatment temperature, determine how long it will take for the nitrogen concentration 10 μm deep into the steel to reach 70% of its value on the surface (i.e., 70% of C_N^*)?

(d) How long would it take the nitrogen concentration 20 μm deep into the steel to reach 70% of its value on the surface?